

Supplementary Material for the Article: Mechanically Verified Optimization and UPPAAL Export for Timed Büchi Automata

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• **ABSTRACT** This supplement records extended implementation and reproduction details for the verified TBA optimization and UPPAAL export pipeline. The Coq preservation results quantify over arbitrary TBAs; measurements are limited to benchmark automata recorded in the project.

S1. IMPLEMENTATION AND REPRODUCTION DETAILS

The verified functions are defined in the local `proof/` tree and extracted into `tool/src/optim.ml`. The project Makefile compiles the Rocq development and extracts the executable functions. Hand-written components include the parser, Spot invocation, LBTT reader, iteration driver, and XML/DOT printers. The script that removes unreachable extracted definitions and the Rocq/OCaml toolchains are part of the implementation trusted base.

The data correspond to five runs per configuration, with a 300-second per-run limit. The machine record is in `evaluation/machine.txt`; raw run files, median summaries, table generator, benchmark formulas, case-study models, and validation scripts are included. Compile instructions are in `README.md`.

The Coq transformations are verified under existential initial clock values. The zero-initialization check used by the formula-generated workflow is not a generic input-format guarantee. Using an arbitrary external TBA requires a separate argument that its initialization and printed representation match the intended UPPAAL semantics.

S2. UPPAAL CASE-STUDY ARTIFACTS

The complete property automata, system model descriptions, queries, and recorded UPPAAL results appear in Section VI of the article. The corresponding UPPAAL models, query files, and outputs are included under `evaluation/case_study/` in the project repository.

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